

Computer Networks

Inter-AS Routing Border Gateway Protocol (BGP)

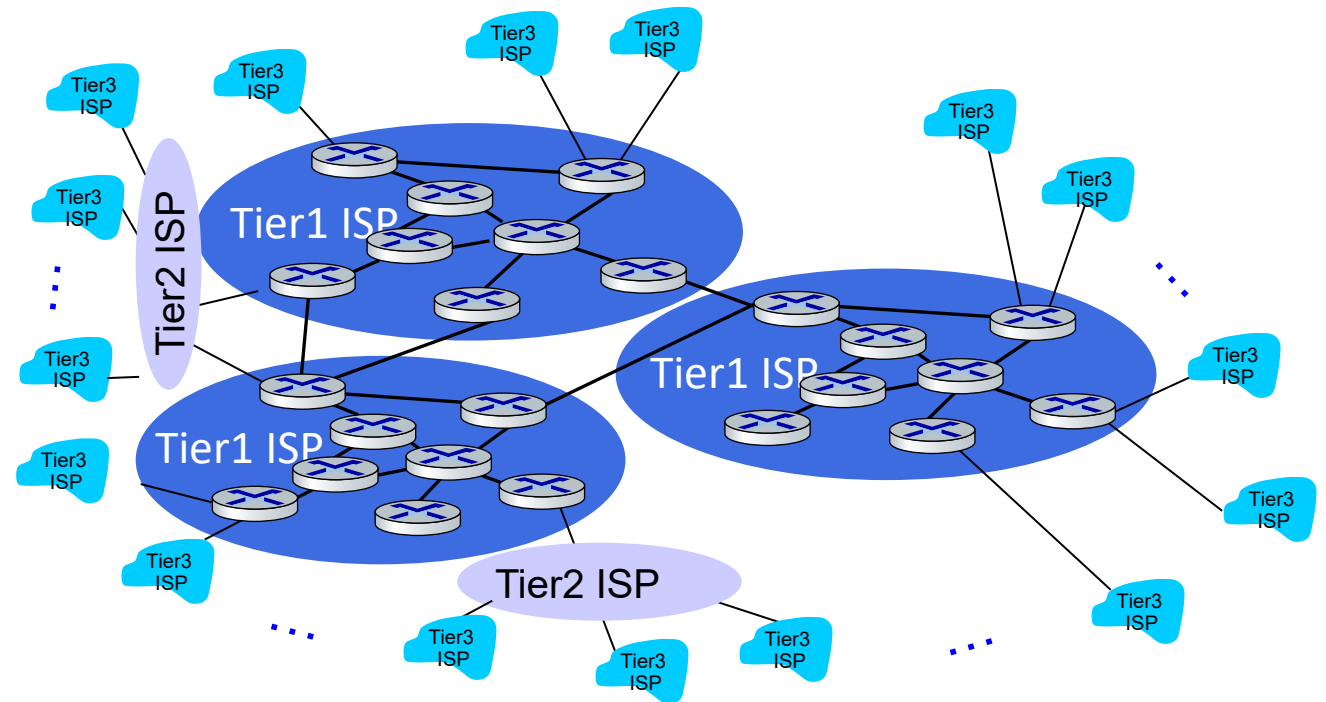
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Inter-AS and Intra-AS Routing

- **AS**: controlled by a single administrative entity, i.e. an ISP

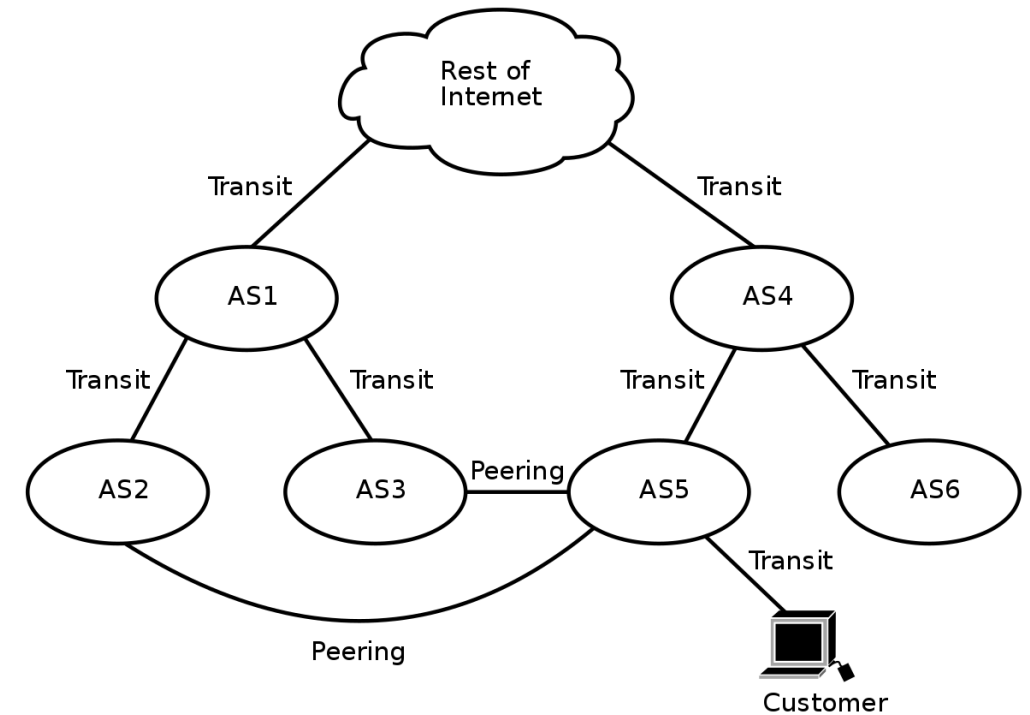


Src:[https://commons.wikimedia.org/wiki/File:India_and_Neighbouring_Countries_Map_\(official_borders\).png](https://commons.wikimedia.org/wiki/File:India_and_Neighbouring_Countries_Map_(official_borders).png)



Inter-AS Routing: BGP

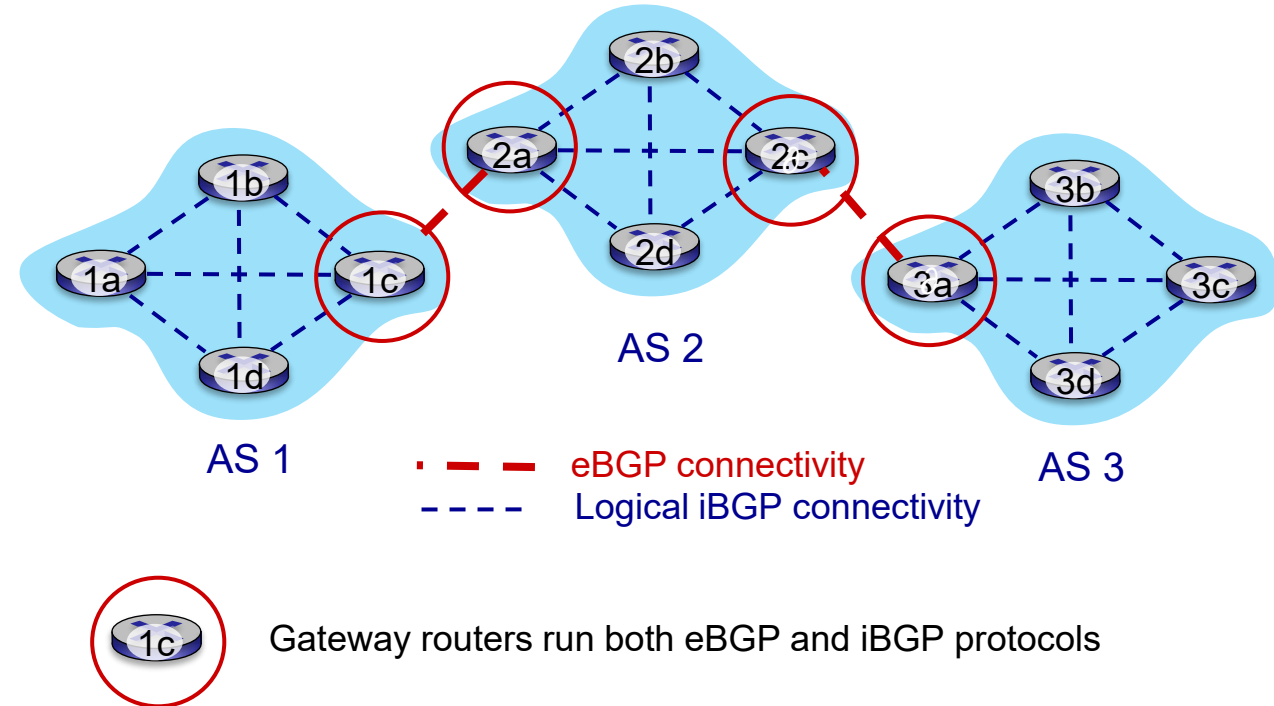
- **BGP (Border Gateway Protocol):** the de facto inter-domain routing protocol
 - “Glue that holds the Internet together”
- Allows subnet to advertise its existence, and the destinations it can reach, to rest of Internet
- Relations among the ASes depend on multiple factors → **competitions, politics, economy** etc.
 - **Transmit:** One AS pays another one for access to the Internet
 - **Peer:** Two ASes exchange traffic between their users freely, and for mutual benefit



Src: <https://fr.wikipedia.org/wiki/Peering#/media/Fichier:AS-interconnection.svg>

Inter-AS Routing: BGP

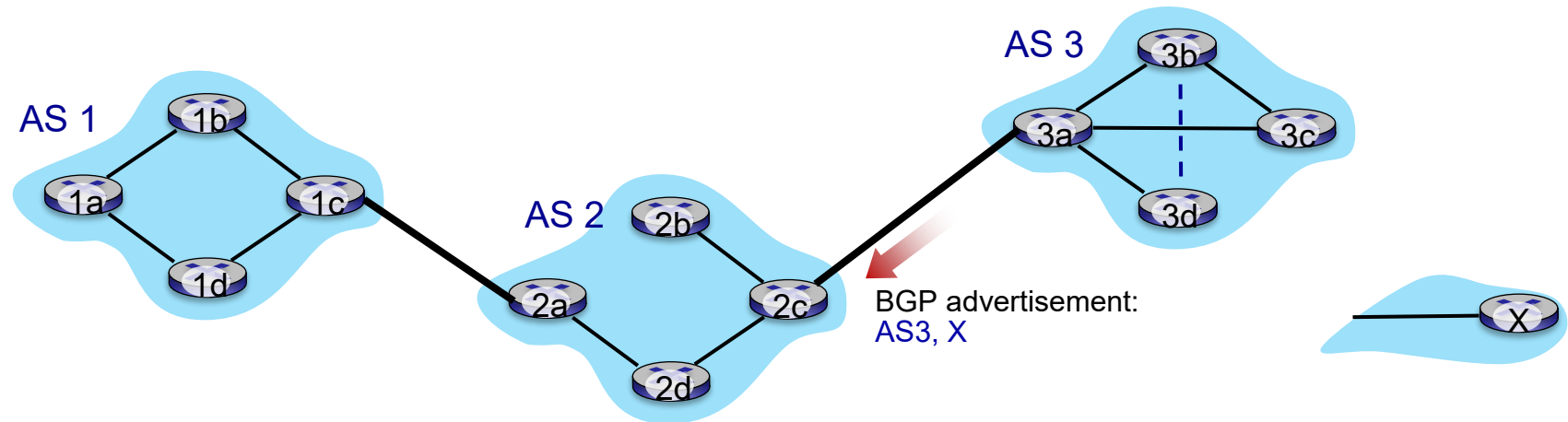
- BGP provides each AS a means to:
 - **eBGP**: obtain subnet reachability information from neighboring ASes
 - **iBGP**: propagate reachability information to all AS-internal routers.
 - Determine “good” routes to other networks based on reachability information and **policy**



BGP and Policy Based Path Advertisement

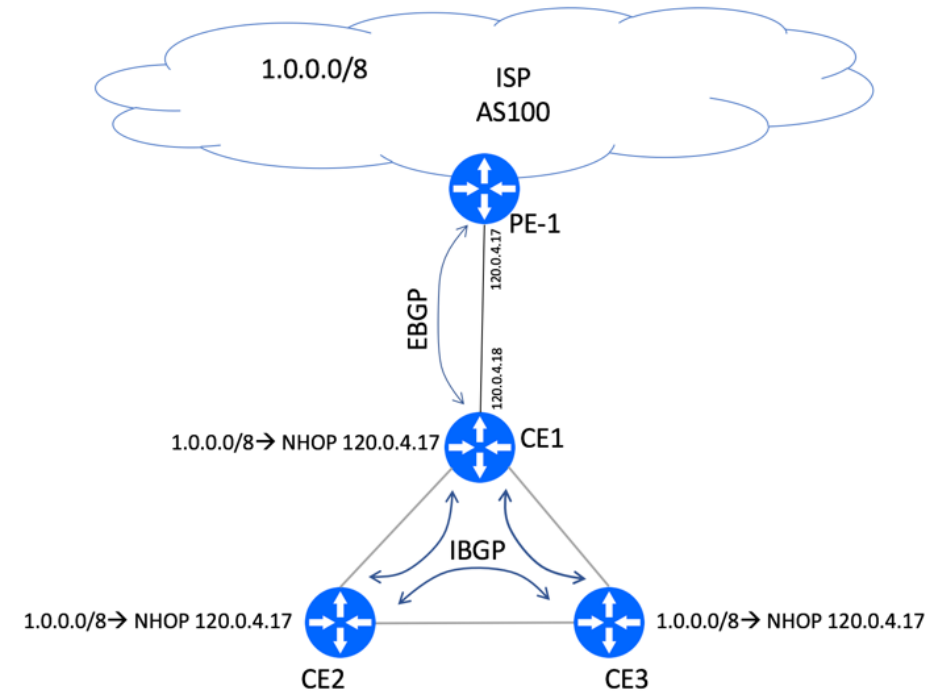
BGP Path Advertisement

- **BGP session**: two BGP routers (“peers”) exchange BGP messages over semi-permanent TCP connection (port 179):
 - Advertising **paths** to different destination network prefixes (BGP is a “**path vector**” protocol)
 - BGP sessions do not correspond to **physical links**
- When AS3 gateway 3a advertises **path AS3,X** to AS2 gateway 2c:
 - AS3 **promises** to AS2 it will forward datagrams towards X



BGP Path Advertisement

- Route advertisement contain an **IP prefix + attributes**
 - Prefix: destination being advertised
 - Two important attributes:
 - **AS-PATH**: list of ASes through which prefix advertisement has passed (**AS50**, **AS76**)
 - **NEXT-HOP**: indicates the router interface that begins the AS-PATH
 - The intra-AS routing protocol will determine the least-cost paths to all subnets attached to the routers in the AS, including the subnet for the link between CE1 and PE-1

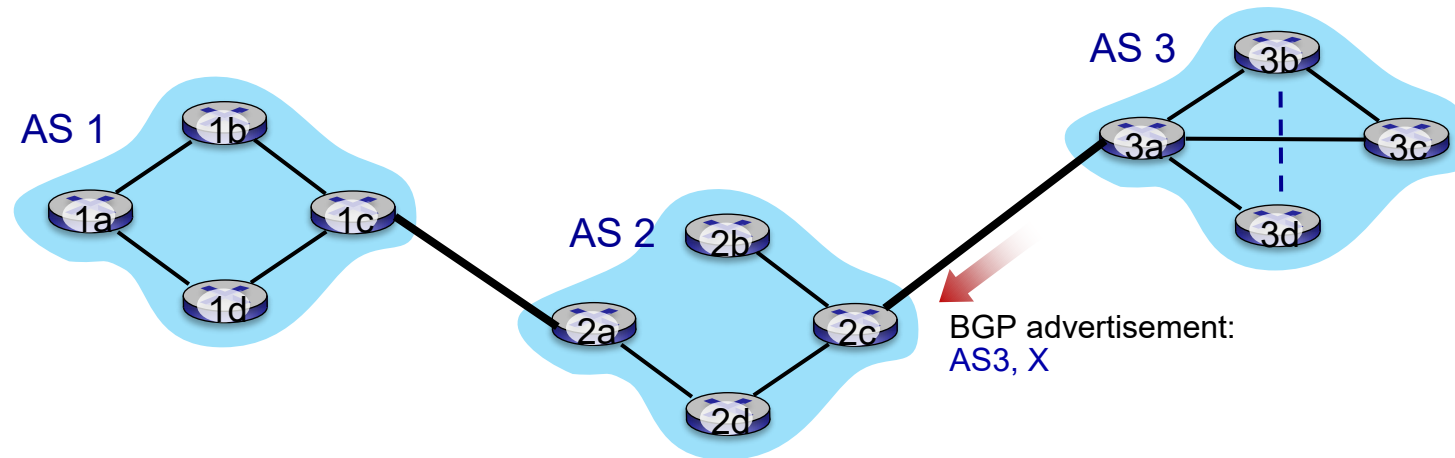


Src: <http://bgphelp.com/2017/03/05/bgp-next-hop-self-explained/>

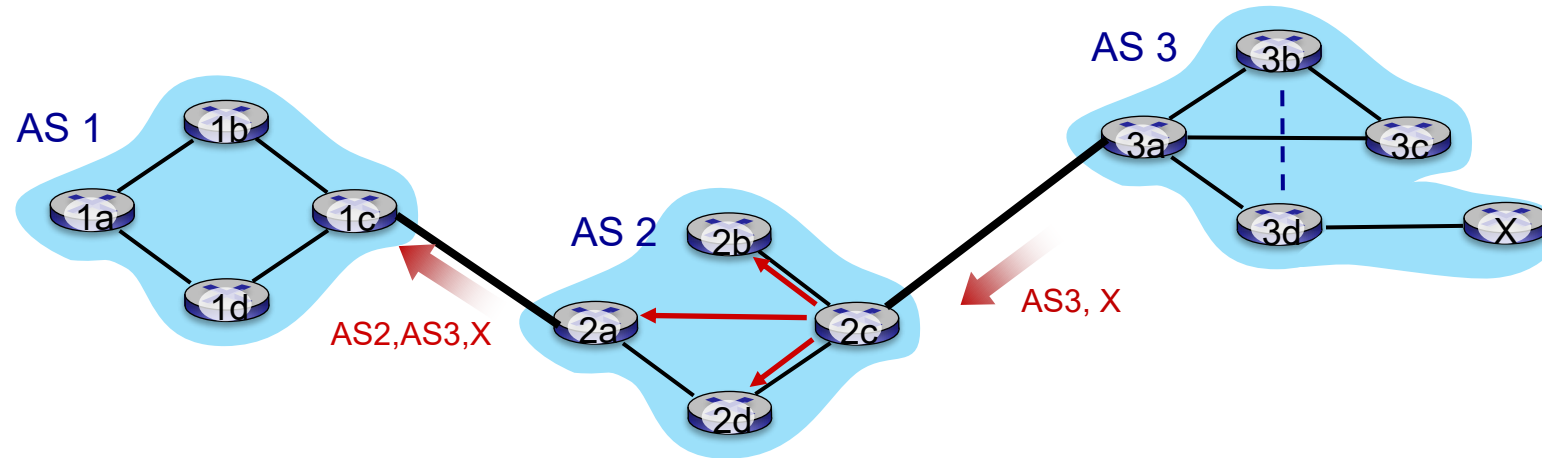
BGP Path Advertisement

■ Policy-based advertisement:

- **Import policy:** Gateway receiving route advertisement uses **import policy** to accept/decline path (e.g., never route through AS Y).
- **Export policy:** AS policy also determines whether to **advertise** path to other neighboring ASes

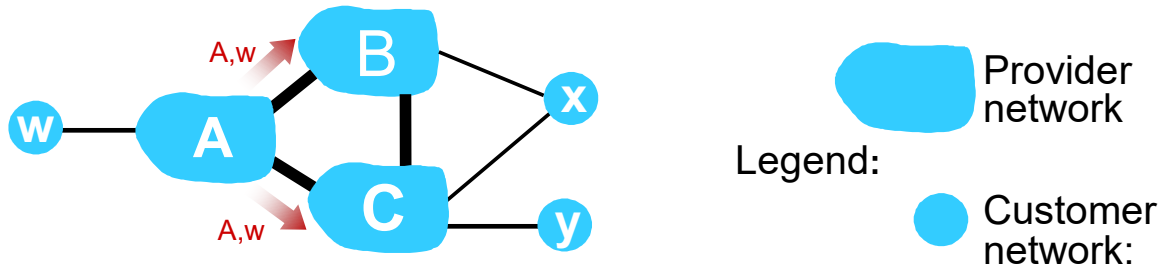


BGP Path Advertisement



- AS2 router 2c receives path advertisement **AS3,X** (via eBGP) from AS3 router 3a
- Based on AS2 policy, AS2 router 2c accepts path AS3,X, propagates (via iBGP) to all AS2 routers
→ **import policy**
- Based on AS2 policy, AS2 router 2a advertises (via eBGP) path **AS2, AS3, X** to AS1 router 1c
→ **export policy**

BGP: Achieving Policy via Advertisements

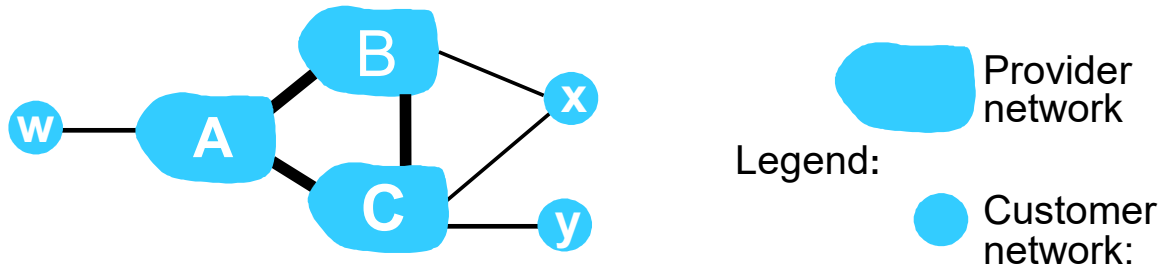


- A,B,C are **provider networks**
- x,w,y are **customer** (of provider networks)
- x is **dual-homed**: attached to two networks

ISP only wants to route traffic to/from its **customer** networks (does not want to carry transit traffic between other ISPs – a typical “real world” policy)

- A advertises path Aw to B and to C
- B **chooses not to advertise** BA_w to C
 - B gets no “revenue” for routing CBA_w, since none of C, A, w are B’s customers
 - C does not learn about CBA_w path
- C will route CA_w (not using B) to get to w

BGP: Achieving Policy via Advertisements



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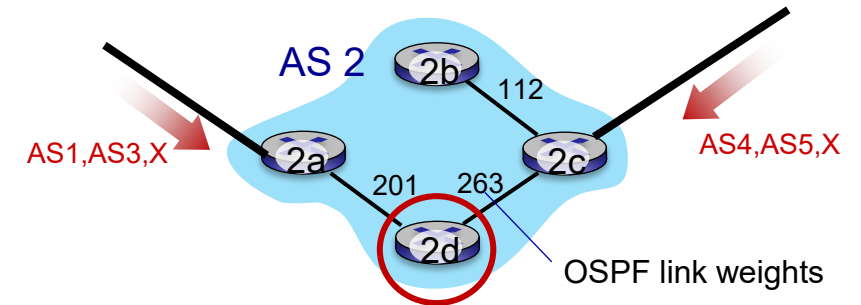
ISP only wants to route traffic to/from its **customer** networks (does not want to carry transit traffic between other ISPs – a typical “real world” policy)

- **Policy to enforce**: x does not want to route from B to C via x
 - So x will not advertise to B a route to C

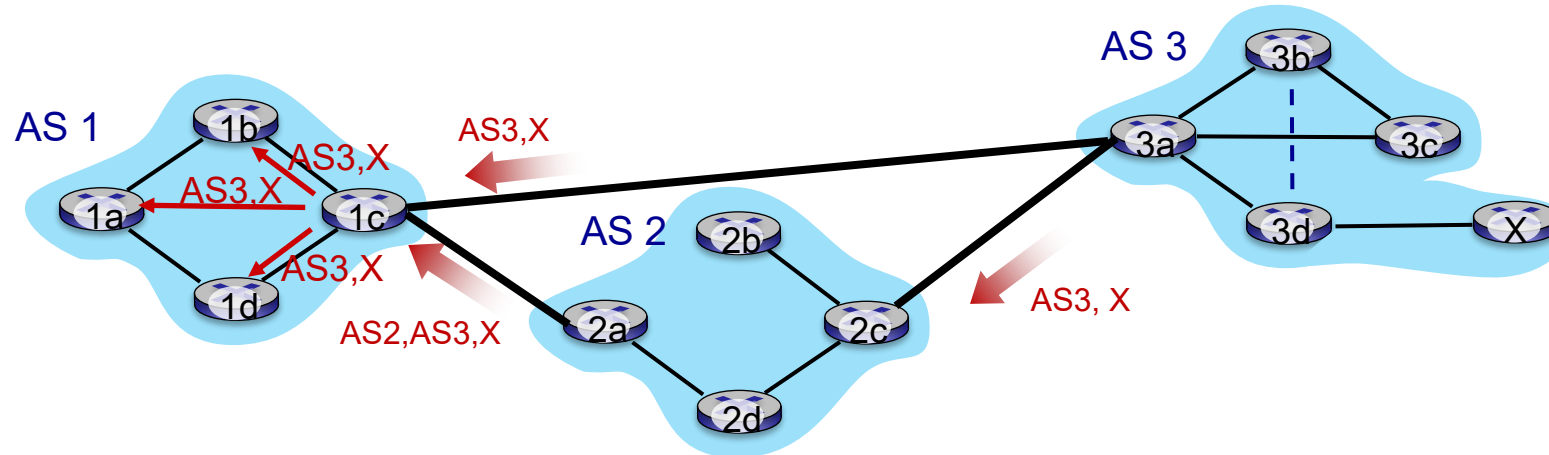
BGP and Policy Based Routing

BGP Route Selection

- Router may learn about more than one route to destination AS, selects route based on:
 - Local **preference** value attribute: policy decision
 - Set by the import policy
 - Shortest **AS-PATH**
 - Included in the route advertisement
 - Closest **NEXT-HOP** router: **Hot potato routing**
 - Based on intra-domain routing protocol (i.e. OSPF)
 - Additional criteria (like smallest NEXT-HOP router id etc.)



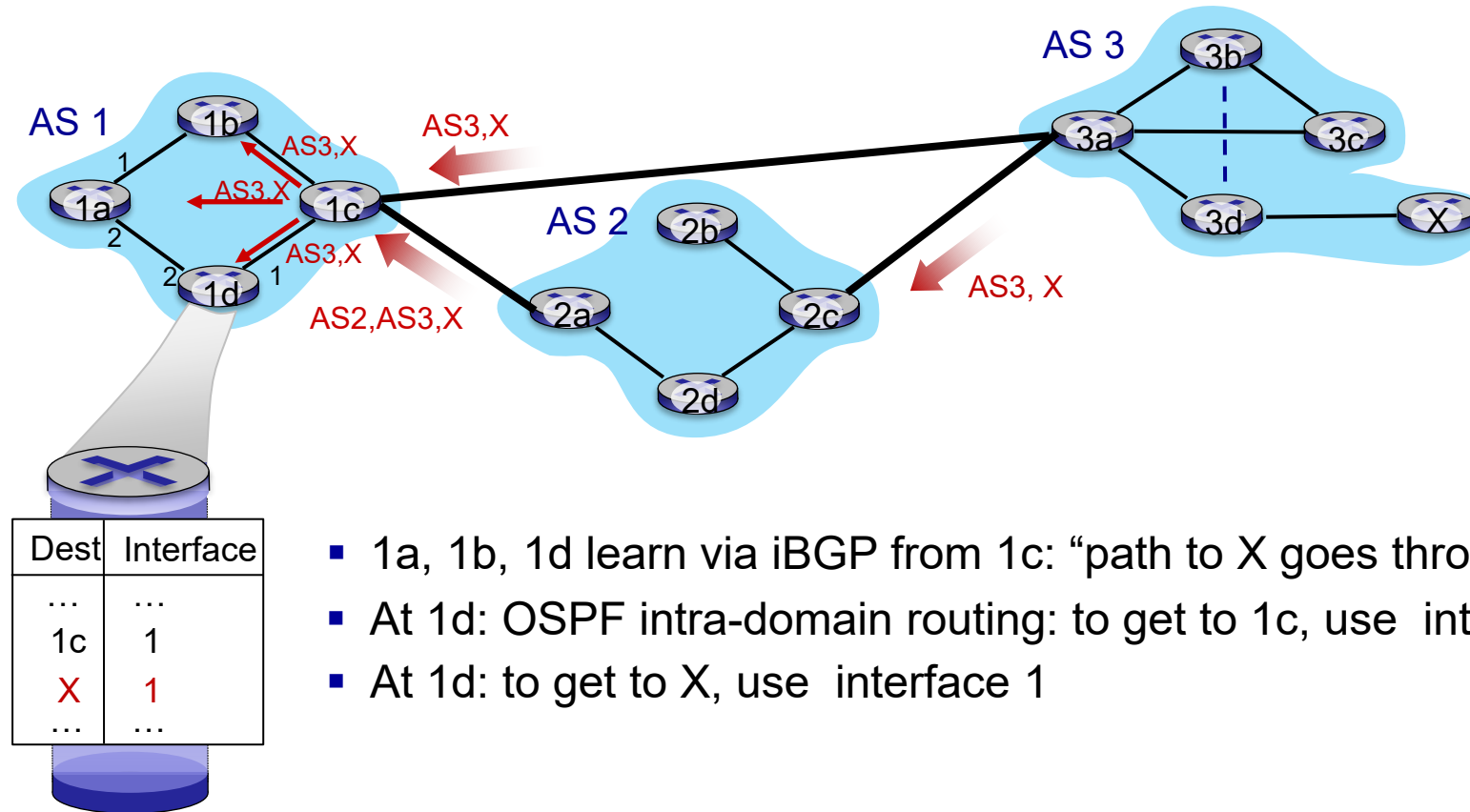
BGP Route Selection



Gateway router may learn about **multiple** paths to destination:

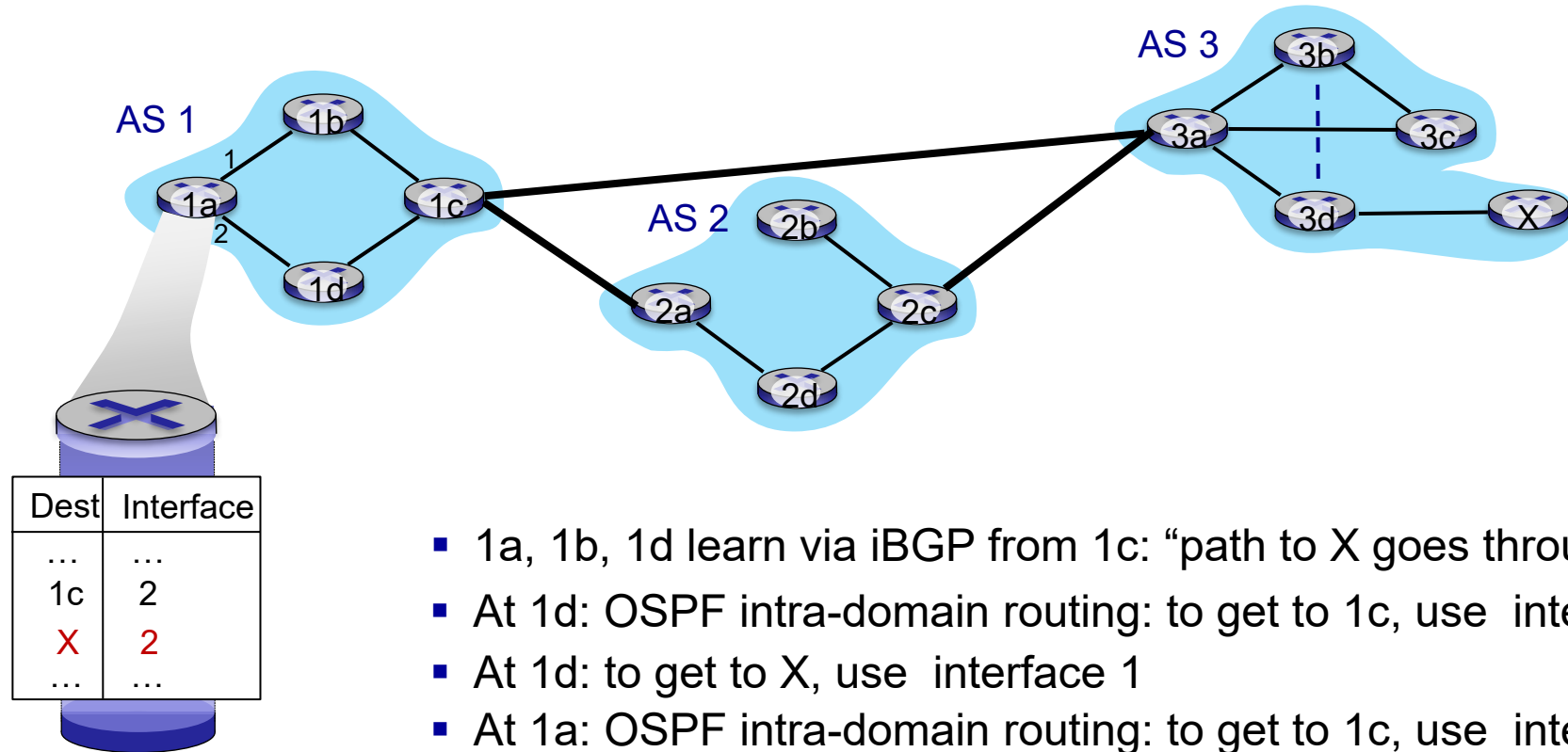
- AS1 gateway router 1c learns path **AS2,AS3,X** from 2a
- AS1 gateway router 1c learns path **AS3,X** from 3a
- Based on **policy**, AS1 gateway router 1c chooses path **AS3,X** and advertises path within AS1 via iBGP

BGP Route Selection



- 1a, 1b, 1d learn via iBGP from 1c: “path to X goes through 1c”
- At 1d: OSPF intra-domain routing: to get to 1c, use interface 1
- At 1d: to get to X, use interface 1

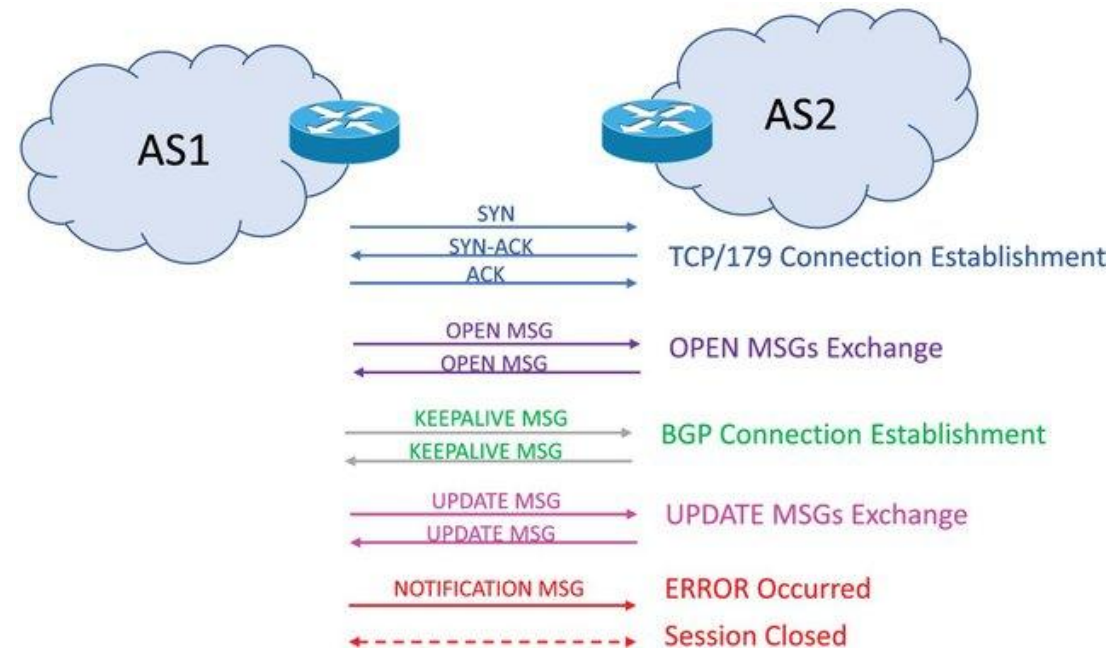
BGP Route Selection



- 1a, 1b, 1d learn via iBGP from 1c: “path to X goes through 1c”
- At 1d: OSPF intra-domain routing: to get to 1c, use interface 1
- At 1d: to get to X, use interface 1
- At 1a: OSPF intra-domain routing: to get to 1c, use interface 2
- At 1a: to get to X, use interface 2

BGP Messages

- BGP messages exchanged between peers over TCP connection
- BGP messages:
 - **OPEN**: opens TCP connection to remote BGP peer and authenticates sending BGP peer
 - **UPDATE**: advertises new path (or withdraws old)
 - **KEEPALIVE**: keeps connection alive in absence of UPDATES; also ACKs OPEN request
 - **NOTIFICATION**: reports errors in previous msg; also used to close connection



Src: <https://onlinelibrary.wiley.com/doi/10.1002/dac.5266?af=R>

Inter-AS vs Intra-AS Routing

- **Inter-AS:** admin wants control over how its traffic routed, who routes through its network
 - **Policy** dominates over performance
 - **Intra-AS:** single admin, so policy less of an issue → can focus on **performance**
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